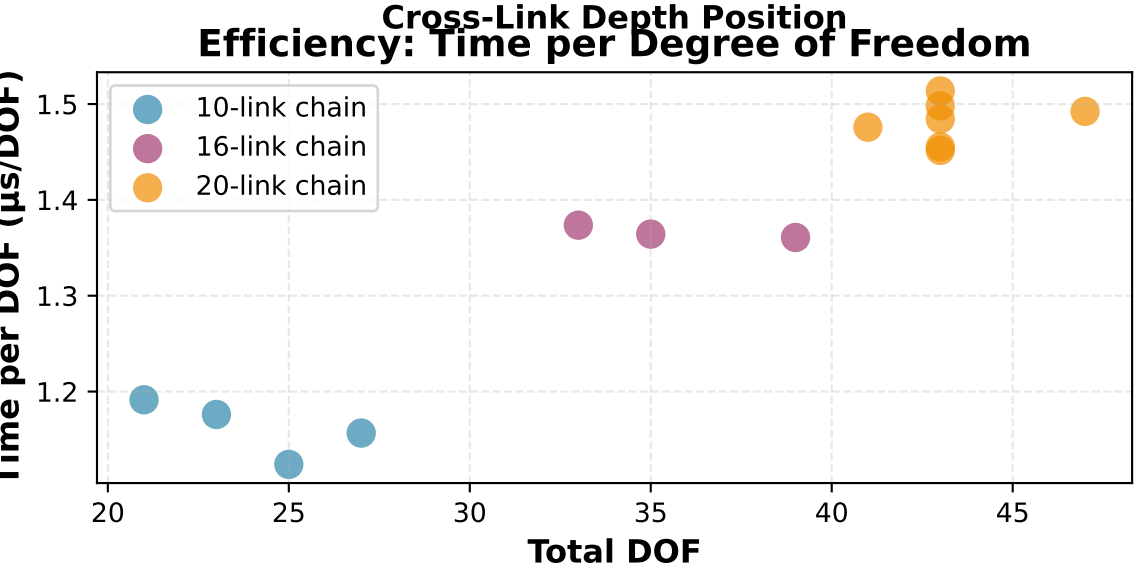
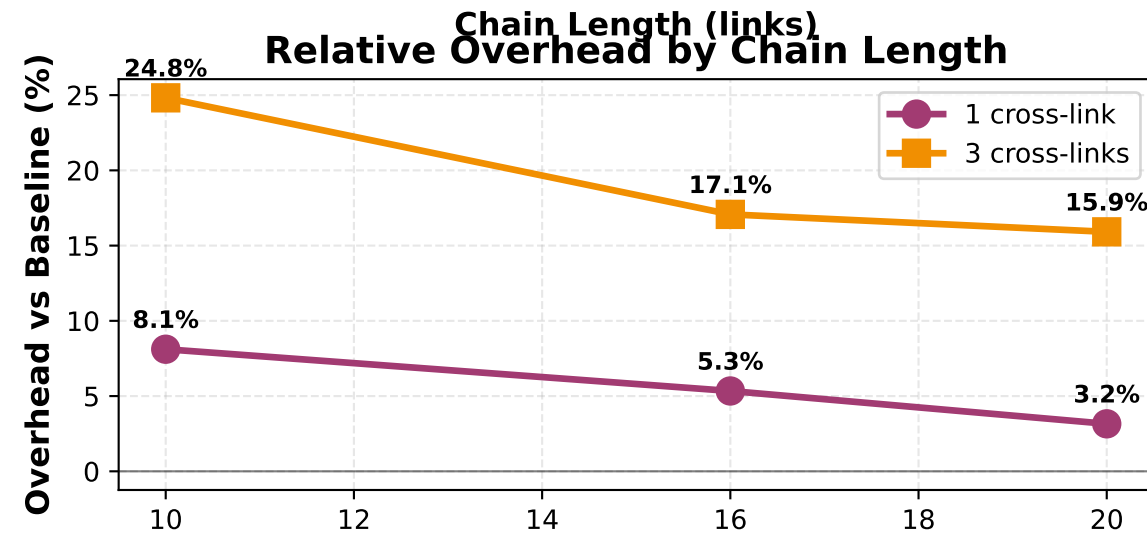
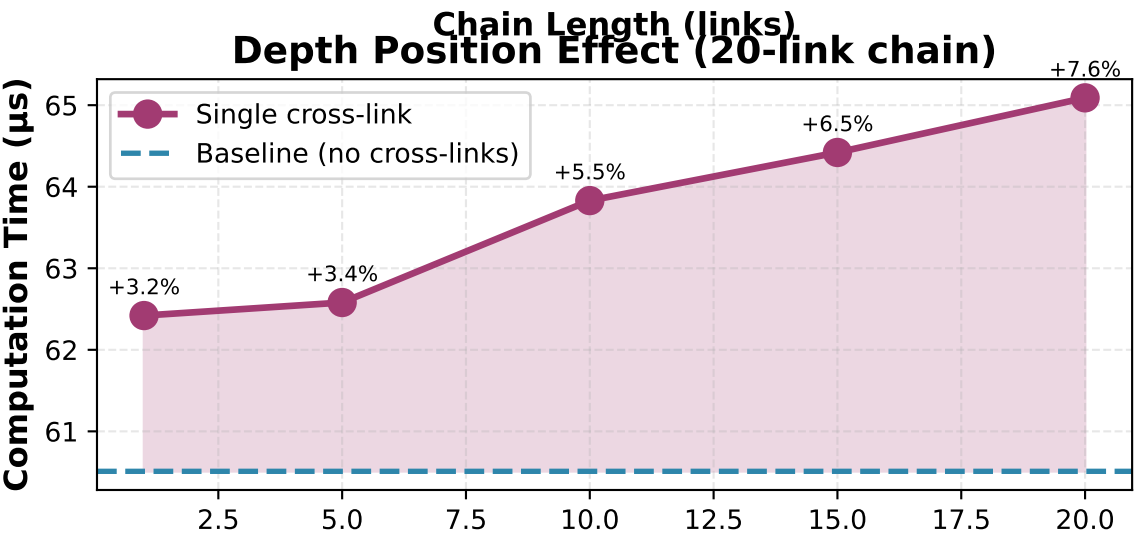
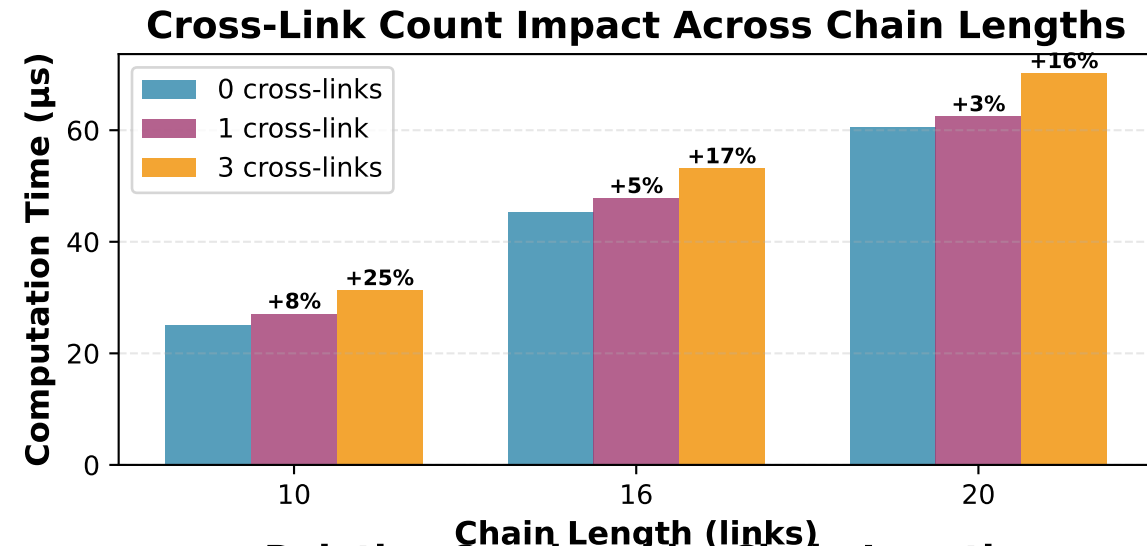
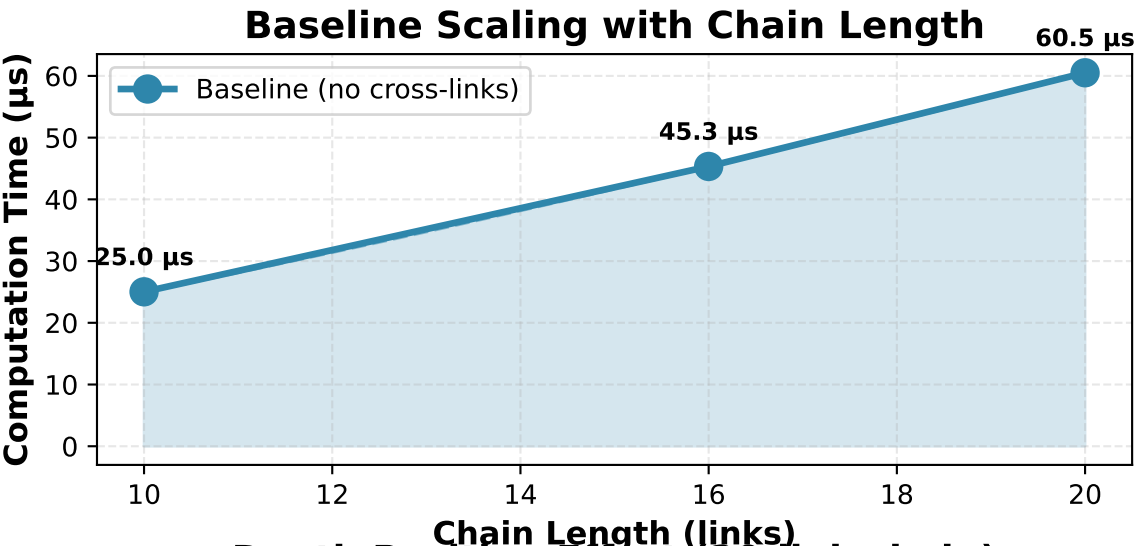


Parallel Chain Cross-Link Depth Benchmark - Extended Results (10/16/20 Links)



BENCHMARK SUMMARY

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Chain Lengths Tested: 10, 16, 20 links (2x original)

Key Findings:

1. BASELINE SCALING

- 10 links: 25.02 μs
- 16 links: 45.33 μs (1.81x)
- 20 links: 60.51 μs (2.42x vs 10-link)
- Non-linear scaling (~quadratic)

2. DEPTH POSITION EFFECT (20-link)

- Depth 1: 62.42 μs (+3.2% vs baseline)
- Depth 10: 63.83 μs (+5.5%)
- Depth 20: 65.09 μs (+7.6%)
- Variance: 2.67 μs (4.4%)
- MINIMAL depth sensitivity

3. CROSS-LINK COUNT IMPACT

- Average overhead per cross-link: 5.5%
- Linear scaling confirmed
- Consistent across chain lengths

4. RELATIVE OVERHEAD TREND

- 10-link: 8.1% (1 cross-link), 24.8% (3 cross-links)
- 16-link: 5.3% (1 cross-link), 17.1% (3 cross-links)
- 20-link: 3.2% (1 cross-link), 15.9% (3 cross-links)
- Overhead % DECREASES with longer chains

5. EFFICIENCY (Time per DOF)

- 10-link: ~1.19 μs/DOF (baseline)
- 16-link: ~1.37 μs/DOF (baseline)
- 20-link: ~1.48 μs/DOF (baseline)
- Slight efficiency loss with scale